EE 121B Fall 2026
Prof. Emaminejad
Homework Assignment 2 Solution

Problem 1: Minority carrier recombination lifetime

Consider a sample of Si doped with 10^{16} cm^{-3} boron, with recombination lifetime 1 μs. It is exposed continuously to light, such that electron-hole pairs are generated throughout the sample at the rate of 10^{20} per cm^3 per second, *i.e.* the generation rate $G_L = 10^{20}/cm^3/s$

a) What are p_0 and n_0?
   $p_0 = 10^{16}\ cm^{-3}$
   $n_0 = 10^{4}\ cm^{-3}$

b) What are Δn and Δp?
   $\Delta n = G_L\tau_n = 10^{20}/cm^3/s \cdot 1\mu s = 10^{14}\ cm^{-3}$
   $\Delta p = \Delta n = 10^{14}\ cm^{-3}$

c) What are p and n?
   $p = p_0 + \Delta p = 10^{16}\ cm^{-3}$
   $n = n_0 + \Delta p = 10^{14}\ cm^{-3}$

d) What is the np product?
   $np = 10^{30}\ cm^{-6}$

Problem 2: Continuity Equation

The two ends of a uniformly doped n-type Si bar of length L are simultaneously illuminated so as to create γN_D excess holes at both x=0 and x=L. The wavelength and intensity of illumination are such that no light penetrates into the interior (0<x<L) of the bar and $\gamma=10^{-4}$. Also, steady state conditions prevail, T=300 K and $N_D >> n_i$.

a) Do low-level injection conditions prevail inside the illuminated bar? Explain. (Lecture 4 slide 18)
   Yes, **low level injection conditions prevail inside the illuminated bar** because the maximum value of excess carrier concentration is $10^{-3}N_D$ which is much less than the equilibrium concentration of majority carriers, *i.e.* $n = n_0 + \Delta n \cong n_0$

b) Write down the differential equation (simplest form possible) you must solve to determine $\Delta p_n(x)$ inside the bar. (Lecture 4 slide 29，30)
   The differential equation to be solved under steady state conditions with no light ($G_L = 0$) inside the bar is

$$D_p \frac{\partial^2 \Delta p_n(x)}{\partial x^2} - \frac{1}{\tau_p}\Delta p_n(x) = 0$$, where D_p is the diffusion coefficient.

c) Write down the general form of the $\Delta p_n(x)$ solution and the boundary condition(s) appropriate for this particular problem. (Lecture 4 slide 31, 32)
   The general solution to this differential equation is

   $\Delta p_n(x) = A\cdot e^{-(x/L_p)} + B\cdot e^{(x/L_p)}$, where $L_p = \sqrt{D_p t_p}$ is the diffusion length.

   The boundary conditions are $\Delta p_n(0) = \gamma N_D$ and $\Delta p_n(L) = \gamma N_D$

d) Establish an expression for the hole current density (J_p) flowing in the illuminated bar at x=0. [Your answer may be left in terms of the arbitrary constant(s) appearing in the general form of the $\Delta p_n(x)$ solution]. (Lecture 4 slide 27)

Assuming that the electric field is small so that the hole drift current density is negligible compared to the hole diffusion current density, *i.e.* $J_p = J_{p,diff}$:

$$J_p\Big|_{x=0} = -qD_p \frac{\partial p_n}{\partial x}\Big|_{x=0} = -qD_p \frac{\partial \Delta p_n}{\partial x}\Big|_{x=0} = -qD_p\left[(-\frac{A}{L_p})e^{-x/L_p} + \frac{B}{L_p})e^{x/L_p}\right]\Big|_{x=0} = -qD_p\left[\frac{B-A}{L_p}\right]$$

Problem 3: Quasi-Fermi Levels

A silicon wafer maintained at T = 300K is doped with 10^{14} cm^{-3} donor atoms.

a) Find the electron and hole concentrations and the location of the Fermi level with respect to the intrinsic Fermi level.

Equilibrium electron concentration is $\boldsymbol{n_0 = N_D = 10^{14}\ cm^{-3}}$; equilibrium hole concentration is $\boldsymbol{p_0 = n_i^2/n = 10^6\ cm^{-3}}$

$(E_F - E_i) = kT\ ln(n/n_i) = 5kT \ln(10) = 4 \times 60$ meV → **E_F is 0.24 eV above the intrinsic Fermi level**.

b) Light irradiating the wafer leads to a steady-state photogenerated density of electrons and holes equal to 10^{12} cm^{-3}. Assume that the wafer is thin when compared to the absorption depth for the light so that the free carriers are uniformly distributed throughout its volume. Find the overall electron and hole concentrations in the wafer and calculate the positions of the quasi-Fermi levels for the two carrier types in an energy band diagram. (Lecture 5 slide 21)

$\Delta p = \Delta n = 10^{12}\ cm^{-3} << n_0$ so $n = n_0 + \Delta n \cong n_0$ (low level injection condition).

Since the electron concentration is not significantly affected, the electron quasi-Fermi level is not changed from the equilibrium Fermi level, *i.e.* **F_N is located 0.24 eV above the intrinsic Fermi level**.

The hole concentration is significantly affected (since $p = p_0 + \Delta p \cong \Delta p$) so the hole quasi-Fermi level is shifted significantly away from the equilibrium Fermi level:

$E_i - F_P = kT\ ln(p/n_i) = 2kT\ ln(10) = 2 \times 60$ meV → **F_P is 0.12 eV below the intrinsic Fermi level.**

c) Repeat the calculations of (b) under the condition that the light intensity is increased so that the photogeneration produces 10^{19} cm^{-3} electron-hole pairs. (Lecture 5 slide 21)

$\Delta p = \Delta n = 10^{19}\ cm^{-3}$ so $n = n_0 + \Delta n \cong \Delta n$ and $p = p_0 + \Delta p \cong \Delta p$ (high level injection condition)

$F_N - E_i$ $= kT\ ln(n/n_i) = 9kT\ ln(10) = 9 \times 60$ meV → **F_N is 0.54 eV above the intrinsic Fermi level.**

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Problem 4: Metal-semiconductor energy-band diagrams

Sketch the energy-band diagrams indicating the Schottky barrier height (Φ_B) and the width of the depletion region (W), for each of the following cases:

a) A contact between NiSi and uniformly doped n-type silicon with $N_D = 10^{17}$ cm^{-3}, $V_A = 0$ V

b) A contact between NiSi and uniformly doped n-type silicon with $N_D = 10^{17}$ cm^{-3}, $V_A = -1$ V

c) A contact between NiSi and uniformly doped n-type silicon with $N_D = 10^{17}$ cm^{-3}, $V_A = 0.2$ V

d) A contact between NiSi and uniformly doped p-type silicon with $N_A = 10^{17}$ cm^{-3}, $V_A = 0$ V

Use the realistic Schottky barrier height values. Assume T=300K.

$\varepsilon_s = 12\times8.85\times10^{-14}$

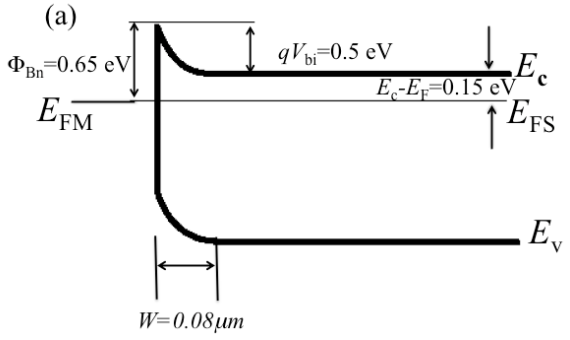
From Lecture 5 Slide 32, one can find the Schottky barrier heights for NiSi contacting Si (Φ_{Bn}=0.65eV and Φ_{Bp}=0.47eV). E_c–E_F in the charge-neutral ("flat-band") region of the semiconductor (away from the depleted region) is determined by using the equation E_c–E_F=kT ln(N_c/N_D) for n-type Si and E_F–E_v =kT ln(N_v/N_A) for p-type Si. The built-in potential is qV_{bi}= Φ_{Bn} – (E_c–E_F) for n-type Si and qV_{bi} = Φ_{Bp} – (E_F–E_v) for p-type material. The width of the depleted region is

$$W = \sqrt{\frac{2\varepsilon_s (V_{bi} - V_A)}{qN}}$$

a) Current is conducted by majority carriers in a Schottky contact. The Schottky barrier height for electrons (Φ_{Bn}=0.65eV) limits electron flow from the metal into n-type Si.

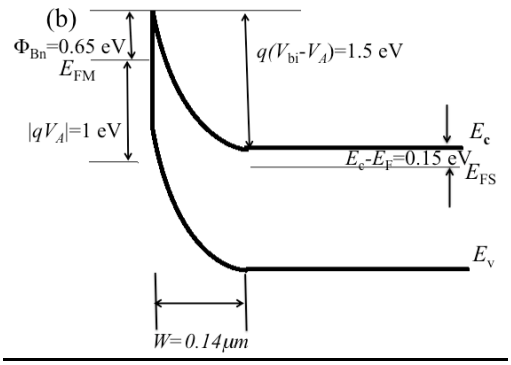
E_c–E_F =0.026·ln(3.2x10^{19}/10^{17})=0.15eV⇒ qV_{bi} = 0.65-0.15=0.5eV

$$W = \sqrt{\frac{2\times12\times8.85\times10^{-14}(0.5)}{1.6\times10^{-19}\times10^{17}}}$$ = 8.14 ×10^{-6} cm= 0.0814μm.



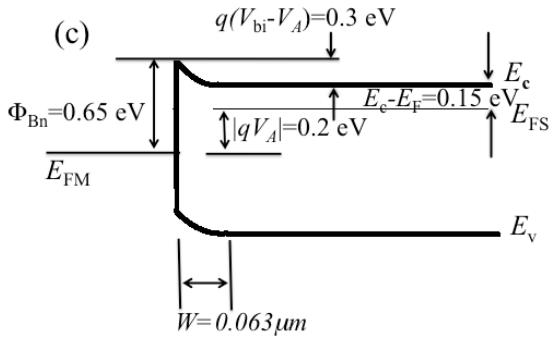
b) Φ_{Bn}, E_c–E_F and V_{bi} are unchanged from (a) since the contacting materials are the same. For an applied voltage equal to -1V the Schottky junction is reverse biased, so the barrier for electron flow from the n-type Si into the metal is increased. (The barrier for electron flow from the metal into the Si is unchanged.) The depletion width is increased (in order to drop a larger voltage across the depleted region):

$$W = \sqrt{\frac{2\times12\times8.85\times10^{-14}(0.5+1)}{1.6\times10^{-19}\times10^{17}}}$$ = 1.4 × 10^{-5} cm= 0.14μm.



c) Φ_{Bn}, E_c–E_F and V_{bi} are unchanged from (a) since the contacting materials are the same. For an applied voltage equal to +0.2V the Schottky junction is forward biased, so the barrier height for electron flow from the n-type Si into the metal is decreased. (The barrier for electron flow from the metal into the Si is unchanged.) The depletion width is decreased (in order to drop a smaller voltage across the depleted region):

$$W = \sqrt{\frac{2\times 12\times 8.85\times 10^{-14}(0.5-0.2)}{1.6\times 10^{-19}\times 10^{17}}} = 6.3\times 10^{-6}\ \text{cm} = 0.063\mu\text{m}.$$



d) . The Schottky barrier height for holes (Φ_{Bp}=0.47eV) limits hole flow from the metal into p-type Si.

E_F–E_v =0.026 *ln*(1.8x10^{19}/10^{17})=0.135eV⇒ qV_{bi} = 0.47-0.135=0.335eV

$$W = \sqrt{\frac{2\times 12\times 8.85\times 10^{-14}(0.33)}{1.6\times 10^{-19}\times 10^{17}}} = 6.6\times 10^{-6}\ \text{cm} = 0.066\mu\text{m}.$$

